



Name: ..... Cohort/Batch: ..... Date: ..... Score: .....

## OPENTOFU PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A DevOps

TOTAL: 10 QUESTIONS

**Q1** What problem does OpenTofu solve in DevOps or infrastructure? **Threat Model**

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**Q6** How does OpenTofu integrate with CI/CD pipelines? **Observability**

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**Q2** What is OpenTofu's core architecture? **Architecture**

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**Q7** How do you debug a failed OpenTofu deployment? **Lifecycle**

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**Q3** How does OpenTofu define infrastructure or configuration as code? **Configuration**

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**Q8** What are security best practices for OpenTofu? **Compliance**

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**Q4** How does OpenTofu ensure idempotency? **Hardening**

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**Q9** How does OpenTofu scale for large environments? **Testing**

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**Q5** How do you handle secrets in OpenTofu? **SSO / IdP**

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**Q10** What are common mistakes teams make when adopting OpenTofu? **Tuning**

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 OpenTofu addresses a specific concern provisioning, Mitigation

OpenTofu addresses a specific concern provisioning, configuration management, orchestration, or CI/CD by automating manual steps that are error-prone, slow, or not reproducible when done by hand.

A6 CI/CD pipelines call OpenTofu in automated steps: Observability

CI/CD pipelines call OpenTofu in automated steps: plan on a pull request to preview changes and apply on merge to main. Approval gates can require a human to review the plan before changes go to production.

A2 OpenTofu has a control plane that stores Primitives

OpenTofu has a control plane that stores desired state and a data plane (agents or push-based SSH) that enforces it. Understanding this distinction clarifies where configuration is evaluated and where changes are applied.

A7 Check OpenTofu's logs for the specific error Automation

Check OpenTofu's logs for the specific error message and the resource that failed. For infrastructure tools, re-run with increased verbosity (-v, --debug). Review the state file or event history to understand what changed before the failure.

A3 OpenTofu uses declarative files (YAML, HCL, or Hardening

OpenTofu uses declarative files (YAML, HCL, or a DSL) to express desired state. A plan/diff phase shows what will change before applying. Version-controlling these files enables peer review and rollback.

A8 Rotate credentials used by OpenTofu, restrict network Governance

Rotate credentials used by OpenTofu, restrict network access to its control plane, enable audit logging of all operations, and use least-privilege service accounts. Never run OpenTofu with admin credentials in production.

A4 OpenTofu operations are designed to produce the Gotchas

OpenTofu operations are designed to produce the same result when run multiple times. Applying the same config twice converges to the desired state rather than creating duplicate resources. This makes automation safe to re-run.

A9 Large OpenTofu deployments use workspaces or namespaces namespaces

Large OpenTofu deployments use workspaces or namespaces to isolate environments, remote state backends for team collaboration, and modular code to avoid a single monolithic config that is slow to plan/apply.

A5 Secrets should never be stored in plain Federation

Secrets should never be stored in plain text in OpenTofu config files. Use a secrets backend (Vault, AWS Secrets Manager) and inject values at runtime via environment variables or encrypted vaults supported by the tool.

A10 Common pitfalls include storing secrets in plain Optimization

Common pitfalls include storing secrets in plain text config, skipping the plan step before apply, not reviewing state drift, and not having a rollback strategy when a deployment fails partway through.